

Conjecturing with Orbit Explorer

TCE 412M · Guest session with Todd Edwards

Predict, run, notice, conjecture: the way mathematicians work on an open problem

ohiomathteacher.github.io/orbit-explorer

AN UNSOLVED PROBLEM

Pick a whole number and apply the rule to the right, over and over. Watch where your number goes.

Where does every starting number end up? Nobody has proved it.

Whatever you notice today is real mathematical evidence, not a rediscovery of something in the back of a book.

THE RULE

If n is even: $n \div 2$

If n is odd: $3n - 1$

Orbit Explorer starts on this rule. Its sliders set m , b , and d in: odd $\rightarrow m \cdot n + b$, even $\rightarrow n \div d$. Today's rule is $m = 3$, $b = -1$, $d = 2$.

COMMIT TO A PREDICTION BEFORE YOU PRESS ANYTHING

1 By hand: 1, 2, 3, 4, 5 ~5 min

Follow each one until something repeats. What happened to each number?

1: _____ 2: _____ 3: _____
4: _____ 5: _____

2 Go bigger, then pool the class data ~10 min

Pick numbers from 21 to 50. Predict, then run each one in Orbit Explorer. Where did it end up? Add each result at loops.mathaiproject.org (the smallest number in its loop is where it ended up). How many different endings has the class found?

START	MY PREDICTION	WHERE IT ENDED UP

3 Change one thing ~5 min

Move b to $+1$, so odd numbers go to $3n + 1$. Set **max steps to 200**. Now run **27**. Before you press: how many steps? How high will it climb?

4 Make a conjecture ~5 min

I conjecture that

I wonder

NOW PUT YOUR TEACHER HAT ON

- What did the tool let you notice that pencil and paper couldn't? What did the pencil give you that the tool didn't?
- Where would you pause a class of seventh graders, and what would you ask?
- Which routine from the field guide could open this lesson *without giving it away*?

Orbit Explorer is free, runs in any browser, and includes an optional AI thinking partner. For more open problems your students can explore: ohiomathteacher.github.io/applet-library/open-problems